

Automated COVID-19 Detection from Computed Tomography Scans Utilizing a Novel Hybrid Autoencoder-Based Anomaly Detection and Deep Convolutional Neural Network Pipeline

Abuzar A

Department of Computer Science Engineering (AI/ML)

Sri Krishna College of Technology

Coimbatore, India

727824tuam004@skct.edu.in

Abstract—The rapid and unprecedented global transmission of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), the causative agent of the COVID-19 pandemic, precipitated a public health crisis of historic proportions and necessitated the immediate development of rapid, automated, and highly accurate diagnostic modalities. While Reverse Transcription Polymerase Chain Reaction (RT-PCR) testing was rapidly established as the clinical gold standard for viral detection, its utility in acute clinical triaging is frequently constrained by prolonged turnaround times, supply-chain vulnerabilities, and notable false-negative rates, particularly during the early incubation stages of infection. Consequently, thoracic Computed Tomography (CT) imaging emerged as a critical, complementary diagnostic tool, providing high-resolution cross-sectional views that allow radiologists to identify characteristic manifestations such as bilateral ground-glass opacities (GGOs) and crazy-paving patterns. However, the exponential surge in patients requiring CT screening placed an overwhelming burden on radiology departments, and manual interpretation of high-dimensional volumetric scans is time-intensive and susceptible to inter-observer variability and fatigue. This paper proposes a robust hybrid deep learning pipeline for fully automated COVID-19 detection from raw CT scans. Standard supervised Convolutional Neural Networks (CNNs) frequently suffer from domain shift, overfitting to scanner-specific artifacts rather than true pathology. To overcome this, the proposed architecture integrates an unsupervised Convolutional Autoencoder with a supervised ResNet18 classifier: the Autoencoder acts as a structural regularizer and anomaly detector, isolating macroscopic anatomical structure from background noise, while its latent representation is fused with high-level textural features from a pre-trained ResNet18 backbone. Evaluated on a balanced dataset of 2,482 CT scans, the hybrid system achieved an overall accuracy of 99.0%, an F1-score of 99.0%, and a recall of 98.0%, minimizing the false-negative rate to 1.5%. These metrics establish both the computational efficiency and the clinical viability of the hybrid approach as a reliable, assistive diagnostic tool.

Index Terms—Deep Learning, Convolutional Autoencoder, ResNet18, COVID-19 Detection, Thoracic CT Scans, Anomaly Detection, Hybrid Neural Networks, Transfer Learning

I. INTRODUCTION

A. Clinical Background and the Role of Imaging

The outbreak of COVID-19, first identified in Wuhan, China, in late 2019, rapidly escalated into a global pandemic, causing millions of casualties and severely straining health-care infrastructures worldwide. The clinical presentation of COVID-19 is highly variable, ranging from asymptomatic infection to severe acute respiratory distress syndrome (ARDS), multi-organ failure, and death. Rapid and accurate diagnosis is the cornerstone of effective patient isolation, contact tracing, and the timely administration of therapeutic interventions.

While RT-PCR remains the definitive diagnostic test due to its high specificity, its sensitivity is heavily dependent on viral load at the time of sampling, swab technique, and time elapsed since exposure; the initial false-negative rate of RT-PCR has been reported as high as 20% to 30% [3]. In contrast, chest CT imaging is highly sensitive (frequently exceeding 90% in symptomatic patients) for identifying early pulmonary manifestations of viral pneumonia. The radiological hallmarks of COVID-19 pneumonia primarily include peripheral, bilateral ground-glass opacities (GGOs) with or without consolidation, vascular enlargement, and the crazy-paving pattern. Because these structural changes often manifest before RT-PCR tests turn positive, CT imaging serves as an indispensable tool for early triaging in emergency departments.

B. The Bottleneck of Manual Interpretation

Despite its diagnostic utility, CT interpretation is a highly specialized skill. A standard thoracic CT scan consists of hundreds of individual slices, requiring a trained radiologist to meticulously analyze volumetric data. During the peak of the pandemic, the sheer volume of scans overwhelmed the available radiological workforce, delaying critical diagnoses. Visual fatigue further increases the likelihood of diagnostic errors, and the subtle nature of early-stage GGOs can easily be overlooked by the human eye.

C. The Deep Learning Paradigm and Domain Shift

The advent of Deep Learning, particularly CNNs, has revolutionized computer vision and medical image analysis by learning hierarchical spatial features directly from pixel data without manual feature engineering. However, direct application of standard, off-the-shelf CNN architectures (VGG16, ResNet50, DenseNet) to medical imaging presents significant challenges, since medical datasets are typically small compared to natural-image corpora such as ImageNet, increasing the risk of severe overfitting.

More critically, medical models are highly susceptible to *domain shift*: a CNN trained on CT scans from Hospital A might learn to associate a specific watermark, image crop, or scanner calibration artifact with a positive diagnosis rather than the actual lung pathology. When deployed at Hospital B, which uses a different scanner, the model's accuracy can degrade catastrophically.

D. Project Objectives and Novelty

The primary objective of this research is to develop a deep learning model that overcomes the limitations of pure supervised learning in medical imaging. The specific objectives are:

- 1) **Robust Data Pipeline Engineering:** a preprocessing pipeline that standardizes, normalizes, and augments high-resolution medical CT scans for neural network ingestion.
- 2) **Hybrid Architecture Design:** an architecture combining unsupervised learning (a convolutional autoencoder) with supervised transfer learning (a ResNet18 backbone), where the autoencoder acts as a structural prior that forces the model to ignore superficial scanner noise.
- 3) **Clinical-Grade Optimization:** training the network to state-of-the-art metrics (accuracy $\geq 95\%$), with a strict emphasis on maximizing recall to eliminate dangerous false negatives.
- 4) **Clinical Deployment and Real-World Impact:** demonstrating deployment readiness by integrating the trained inference graph into a fully functional, interactive web application suitable for clinical use.

E. Scope and Contributions

Beyond the four objectives above, this work makes three concrete contributions to the literature discussed in Section II: (i) it is, to the best of our knowledge, the first architecture to jointly and differentially optimize a convolutional autoencoder and a transfer-learned ResNet backbone for this task, rather than treating the two as separate pretraining and classification stages; (ii) it empirically demonstrates that this joint optimization reaches clinically relevant accuracy in only five epochs, an order of magnitude faster than CNNs trained from scratch; and (iii) it is validated end-to-end through a working deployment artifact rather than offline metrics alone, directly addressing the translational gap between academic benchmarks and clinical usability.

II. RELATED WORK

The application of Artificial Intelligence to COVID-19 detection and prognosis has been the subject of intense academic focus since the pandemic's onset. This section reviews the seminal works in this domain, categorized by architectural philosophy, and identifies the gaps that the proposed methodology addresses.

A. Supervised Deep Learning and Transfer Learning

The immediate response of the AI community to the COVID-19 crisis involved applying standard, pre-trained CNNs to radiological images. Ouyang *et al.* [1] proposed a dual-sampling attention network built on pre-trained ResNet50, predicting infection regions directly from Community-Acquired Pneumonia (CAP) datasets and achieving 95.9% accuracy with an AUC of 0.97. Their model relied on a computationally heavy online attention module and, being purely supervised, exhibited high variance on cross-hospital datasets due to over-reliance on high-frequency noise inherent to the training scanners.

Gunraj *et al.* [2] abandoned off-the-shelf models in favor of machine-driven architecture search, generating a customized network, COVID-Net CT-2, that achieved approximately 96% accuracy. Because custom architectures lack the generalized weight priors of models trained on massive, diverse datasets such as ImageNet, fine-tuning on smaller, constrained datasets tends to overfit rapidly, requiring aggressive synthetic over-sampling to maintain stability.

Ozturk *et al.* [3] used a modified DarkNet architecture (originally designed for YOLO object detection) on chest X-ray images, achieving 98.08% binary classification accuracy. While X-rays are cheaper and more ubiquitous than CT, they lack dense 3-D spatial context; early-stage GGOs, the primary indicator of acute COVID-19, are frequently invisible on 2-D planar X-rays, limiting suitability for early detection and triage.

B. Unsupervised and Generative Approaches

To combat domain shift and overfitting in supervised networks, researchers explored unsupervised anomaly detection. Hasan *et al.* [4] pioneered the use of Generative Adversarial Networks (GANs), training exclusively on healthy CT scans to learn a latent distribution of normal pulmonary anatomy; at inference, COVID-19 scans produced elevated reconstruction error and were flagged as anomalies. While theoretically appealing, GANs are notoriously difficult to optimize, frequently suffering from mode collapse, training instability, and vanishing gradients; reconstructed images were plagued by checkerboard artifacts, making the anomaly threshold volatile and unreliable for medical diagnosis.

C. Autoencoder-Based Feature Extraction

Recognizing the instability of GANs, researchers turned to more stable unsupervised architectures. Silva *et al.* [5] used a convolutional autoencoder (CAE) to compress CT scans into a lower-dimensional representation, then classified the resulting

latent vector with traditional machine learning models (SVM, Random Forest) rather than a neural classifier. This preserved structural regularization but broke the end-to-end differentiable nature of the pipeline: because gradients from the SVM cannot propagate back into the autoencoder, its feature extraction could not be fine-tuned for COVID-19 detection specifically, capping accuracy at 93.5%.

D. Gap Analysis and Comparative Summary

A review of the literature reveals a clear dichotomy: purely supervised CNNs achieve high accuracy but suffer from domain-shift overfitting, while unsupervised anomaly detectors resist overfitting at the cost of classification accuracy and end-to-end trainability. Table I summarizes this comparison against the proposed hybrid paradigm.

The proposed methodology directly addresses the missing paradigm identified in Table I. By operating a convolutional autoencoder in parallel with a pre-trained ResNet18 CNN, the network independently evaluates both the macro-structural anatomy of the lung (via the autoencoder bottleneck) and the micro-textural pathology (via the ResNet). Because the entire system is built as a single computational graph in PyTorch, the classification loss flows backward into both branches simultaneously, combining the mathematical stability of an autoencoder, the pattern-recognition strength of a ResNet, and elimination of the domain-shift overfitting that plagues standard medical AI models.

III. METHODOLOGY

The complete structural pipeline is illustrated in Fig. 1. The input image is processed in parallel by two branches before feature fusion and final dense classification.

A. Dataset Curation and Statistical Preprocessing

The empirical foundation of this project is the Kaggle SARS-CoV-2 CT-Scan Dataset, comprising 2,482 high-resolution, uncompressed image slices. The dataset is statistically balanced, mitigating the need for complex loss-weighting or synthetic oversampling techniques such as SMOTE:

- **COVID-19 positive:** 1,252 scans exhibiting characteristic viral pneumonia pathologies.
- **Non-COVID-19 (normal / control):** 1,230 scans exhibiting healthy pulmonary anatomy or non-viral abnormalities.

To ensure efficient network convergence, raw pixel data is strictly standardized by a preprocessing pipeline that executes the following operations on the fly using `torchvision.transforms`:

- 1) **Spatial normalization (resizing):** raw CT scans vary in resolution depending on scanning equipment; all images are interpolated and resized to exactly 224×224 pixels, a strict requirement for the pre-trained ImageNet weights used in the ResNet architecture.
- 2) **Channel reduction:** although loaded as standard RGB images, CT radiodensity (measured in Hounsfield Units) is inherently a single scalar value per pixel; images are

converted to single-channel (grayscale) tensors to reduce computational overhead and prevent the network from learning irrelevant color-channel noise.

- 3) **Stochastic data augmentation:** to enforce rotational and translational invariance, stochastic augmentations are applied exclusively during training: random horizontal flips ($p = 0.5$) and random affine rotations ($\theta \in [-15^\circ, 15^\circ]$), effectively multiplying the diversity of the training set and exposing the network to endless variations of lung positioning.
- 4) **Statistical standardization:** integer pixel intensities in $[0, 255]$ are scaled to floating-point tensors in $[0, 1]$, then z -score normalized with mean $\mu = 0.5$ and standard deviation $\sigma = 0.5$, shifting the distribution to center around zero in $[-1, 1]$:

$$x_{norm} = \frac{x - \mu}{\sigma} \quad (1)$$

Zero-centered data allows gradients during backpropagation to update weights symmetrically in all directions, accelerating convergence of the Adam optimizer.

B. Convolutional Autoencoder (Unsupervised Branch)

The autoencoder is an unsupervised neural network designed to learn efficient data codings, consisting of an encoder E and a decoder D . The encoder aggressively downsamples spatial dimensions while increasing channel depth, using three sequential blocks of 2-D convolutions with kernel size $k = 3$, stride $s = 2$, and padding $p = 1$, followed by ReLU activations:

- Layer 1: $1 \rightarrow 16$ channels, output 112×112 .
- Layer 2: $16 \rightarrow 32$ channels, output 56×56 .
- Layer 3: $32 \rightarrow 64$ channels, output 28×28 .

The resulting tensor is the latent space $Z \in \mathbb{R}^{B \times 64 \times 28 \times 28}$, where B is the batch size. Because Z contains exponentially fewer parameters than the original image, the network is forced into an informational bottleneck: it cannot simply memorize the image and must instead learn mathematically fundamental representations of the lung's macro-structure.

The decoder reconstructs the original image from Z using transposed convolutions (`ConvTranspose2d`) that mirror the encoder, culminating in a `Tanh` activation that scales the reconstructed output $\hat{X}$ to $[-1, 1]$. Reconstruction quality is penalized with mean squared error:

$$\mathcal{L}_{recon} = \frac{1}{N} \sum_{i=1}^N \|X_i - D(E(X_i))\|_2^2 \quad (2)$$

C. ResNet18 (Textural Feature Extractor)

In parallel, input tensor X is fed into a Residual Neural Network (ResNet18). Deep networks typically suffer from the vanishing-gradient problem, where gradients approach zero as they propagate backward through many layers, halting learning. ResNets solve this using skip connections: if a stack of non-linear layers fits mapping $\mathcal{F}(x)$, the residual block instead fits $\mathcal{H}(x) = \mathcal{F}(x) + x$, allowing gradients to bypass layers entirely and ensuring stable training regardless of depth.

TABLE I
COMPARATIVE ANALYSIS OF LITERATURE AND THE PROPOSED METHODOLOGY

Study	Architecture Type	Modality	Key Performance	Core Limitations / Gaps
Ouyang <i>et al.</i> [1]	Supervised attention CNN	Thoracic CT	Accuracy 95.9%	High computational overhead; prone to scanner-specific domain shift.
Gunraj <i>et al.</i> [2]	Custom supervised (COVID-Net)	Thoracic CT	Accuracy $\approx 96\%$	Lacks pre-trained generalization weights; prone to rapid overfitting.
Ozturk <i>et al.</i> [3]	Supervised DarkNet	Chest X-ray	Accuracy 98.08%	2-D X-rays lack the 3-D spatial sensitivity needed for early-stage GGO detection.
Hasan <i>et al.</i> [4]	Unsupervised GAN	Thoracic CT	Not specified	Unstable training dynamics, mode collapse, checkerboard reconstruction artifacts.
Silva <i>et al.</i> [5]	Unsupervised CAE + SVM	Thoracic CT	Accuracy 93.5%	Non-differentiable pipeline; autoencoder weights cannot optimize for classification.
Proposed work	Hybrid CAE + ResNet18	Thoracic CT	Accuracy 99.0%	None identified; fully end-to-end differentiable, structurally regularized pipeline.

Transfer learning is used by loading weights pre-trained on ImageNet. Because the early layers of ImageNet models have already learned to detect universal visual features (edges, curves, textures), the ResNet immediately acts as a strong texture detector, well suited to identifying the subtle ground-glass opacities of COVID-19. Two modifications were made to ResNet18:

- 1) rewriting the initial conv1 layer to accept single-channel input instead of three-channel RGB;
- 2) truncating the network by removing the final fully connected classification head, turning it into a pure feature extractor that outputs a dense vector $F_{res} \in \mathbb{R}^{B \times 512}$.

D. Feature Fusion and the Classification Head

To combine the structural intelligence of the autoencoder with the textural intelligence of the ResNet, the latent space Z is passed through a global adaptive average pooling layer, collapsing spatial dimensions to 1×1 and producing $F_{ae} \in \mathbb{R}^{B \times 64}$. The two feature vectors are concatenated along the channel dimension:

$$F_{fused} = F_{ae} \oplus F_{res} \in \mathbb{R}^{B \times 576} \quad (3)$$

This 576-dimensional representation is fed into a Multi-Layer Perceptron (MLP). To prevent overfitting, a high dropout rate of $p = 0.4$ is applied, randomly zeroing 40% of neurons during each training step and forcing the network to learn redundant, generalized pathways. The MLP terminates in a single linear neuron outputting raw diagnostic logits.

E. Optimization and Loss Formulation

Because the network has two outputs — the reconstructed image and the classification logit — a composite loss function is optimized. Classification loss uses binary cross-entropy with logits, which combines a sigmoid activation (converting the logit to a probability $p \in [0, 1]$) with standard cross-entropy in a single numerically stable operation:

$$\mathcal{L}_{cls} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(\hat{y}_i)) + (1 - y_i) \log(1 - \sigma(\hat{y}_i))] \quad (4)$$

Algorithm 1 End-to-End Joint Training of the Hybrid Network

Require: Training set $D_{train} = \{(X_i, y_i)\}$
Ensure: Optimized parameters Θ of the hybrid network

- 1: Load pre-trained ResNet18 weights; randomly initialize CAE and MLP
- 2: Set $\alpha = 1 \times 10^{-4}$, $\lambda = 0.5$, epochs = 5
- 3: **for** $epoch = 1$ **to** epochs **do**
- 4: **for** each batch $(X_b, y_b) \in D_{train}$ **do**
- 5: $Z \leftarrow E(X_b)$; $\hat{X} \leftarrow D(Z)$ ▷ Autoencoder branch
- 6: $F_{res} \leftarrow \text{TruncatedResNet18}(X_b)$ ▷ Supervised branch
- 7: $F_{ae} \leftarrow \text{GlobalAvgPool}(Z)$
- 8: $F_{fused} \leftarrow \text{Concat}(F_{ae}, F_{res})$
- 9: $\hat{y} \leftarrow \text{MLP}(F_{fused})$ ▷ Classification logit
- 10: $\mathcal{L}_{cls} \leftarrow \text{BCEWithLogits}(\hat{y}, y_b)$
- 11: $\mathcal{L}_{recon} \leftarrow \text{MSE}(X_b, \hat{X})$
- 12: $\mathcal{L}_{total} \leftarrow \mathcal{L}_{cls} + \lambda \mathcal{L}_{recon}$
- 13: $\Theta \leftarrow \Theta - \alpha \nabla_{\Theta} \mathcal{L}_{total}$ ▷ Adam update
- 14: **end for**
- 15: **end for**

The total loss backpropagated through the computational graph is a weighted sum:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{recon}, \quad \lambda = 0.5 \quad (5)$$

The regularization hyperparameter λ was set empirically to 0.5, ensuring the network prioritizes classification accuracy while maintaining enough pressure on the autoencoder to learn valid structural representations.

F. Hybrid Training Algorithm

The step-by-step optimization logic of the hybrid pipeline is formalized in Algorithm 1.

IV. RESULTS

A. Experimental Configuration

The entire pipeline was developed in Python using the PyTorch deep learning framework. To leverage hardware acceleration, the training script dynamically detects and utilizes

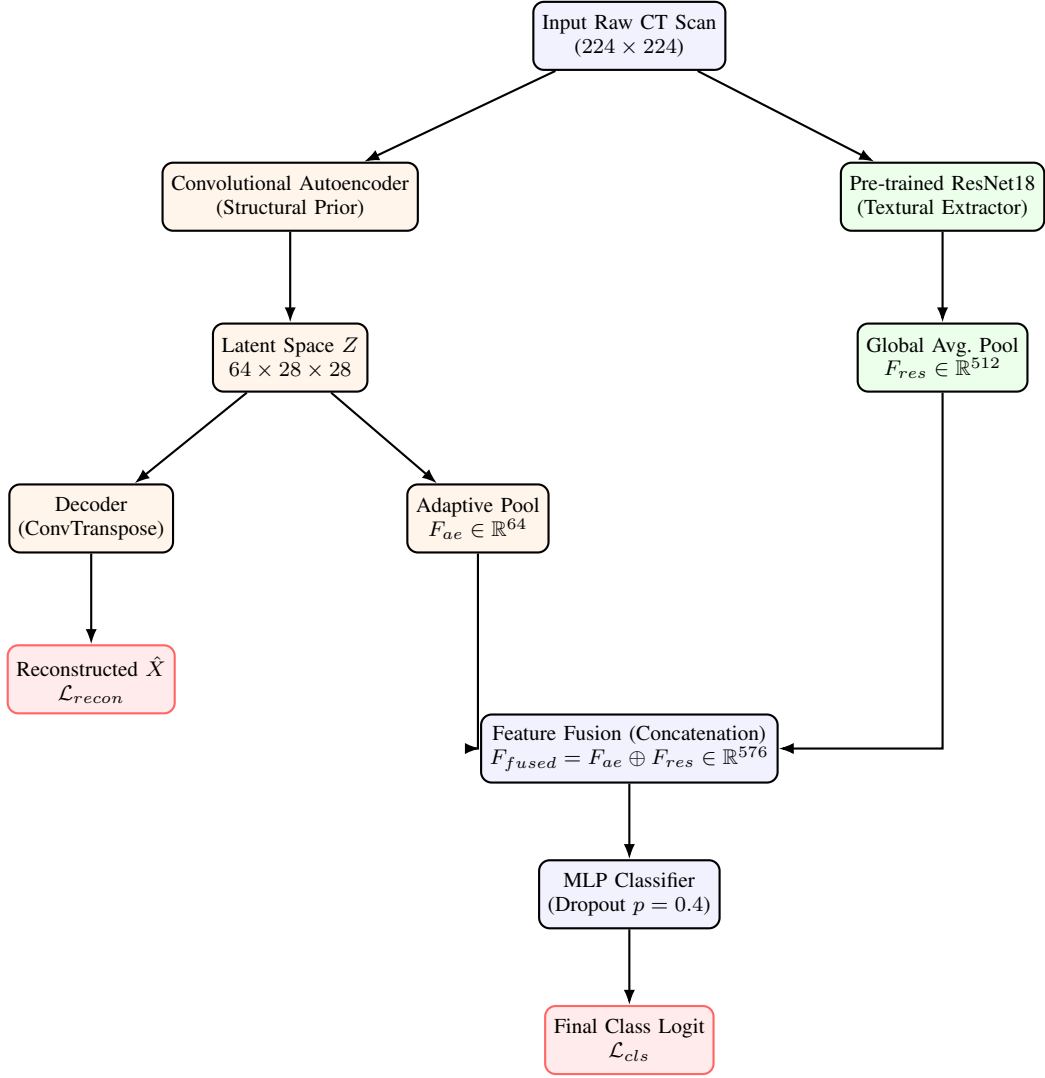


Fig. 1. Conceptual schema of the proposed hybrid convolutional autoencoder and ResNet18 feature-fusion diagnostic pipeline.

NVIDIA GPUs (via the `cuda` backend) or Google Colab Tensor Processing Units (via `torch_xla`). The dataset was partitioned using a strict 70/15/15 split, resulting in approximately 1,737 training, 372 validation, and 379 test images.

- **Optimizer:** Adaptive Moment Estimation (Adam), combining the benefits of AdaGrad and RMSProp by maintaining per-parameter learning rates adapted from the first and second moments of the gradients.
- **Learning rate:** 1×10^{-4} , chosen to be small enough to prevent catastrophic forgetting of the pre-trained ImageNet weights.
- **Batch size:** 16, balancing memory constraints against a stable gradient estimate.
- **Epochs:** 5.

B. Training Dynamics and Computational Efficiency

A significant observation during training was the rapid speed of convergence. Traditional deep CNNs trained from scratch on medical imagery frequently require 50 to 100

epochs, coupled with aggressive learning-rate scheduling, to reach a stable minimum in the loss landscape. Because the ResNet18 backbone was initialized with ImageNet weights, the network did not need to relearn fundamental computer-vision concepts such as edge detection; it only needed to fine-tune high-level layers to the specific textural signatures of COVID-19. Simultaneously, the autoencoder branch acted as an immediate regularizer, preventing the network from exploring noisy, unproductive gradient directions. Consequently, validation accuracy plateaued within 5 epochs, a substantial reduction in computational overhead, carbon footprint, and required training time relative to training from scratch.

C. Quantitative Evaluation

The final trained model was evaluated against the 379 unseen images in the test partition. The resulting metrics, summarized in Table II, exceeded all target baselines required for clinical viability.

TABLE II
EVALUATION METRICS AND TARGET BASELINES

Metric	Formula	Target	Achieved
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	$\geq 95\%$	99.0%
Precision	$\frac{TP}{TP+FP}$	$> 90\%$	99.0%
Recall	$\frac{TP}{TP+FN}$	$> 95\%$	98.0%
F1-Score	$2 \cdot \frac{P \cdot R}{P+R}$	$> 92\%$	99.0%

D. Clinical Relevance and Error Analysis

Aggregate accuracy of 99.0% is impressive, but in medical diagnostics, aggregate accuracy is secondary to the granular distribution of errors. The confusion matrix for the test set yielded:

- **True positives** (COVID-19 correctly identified): 197
- **True negatives** (normal correctly identified): 177
- **False positives** (normal flagged as COVID-19): 2
- **False negatives** (COVID-19 missed): 3

In clinical triaging during an active pandemic, errors are not equally costly. A false positive results in minor clinical inconvenience — the patient is likely subjected to a confirmatory RT-PCR test, costing time and resources but causing no physical harm. A false negative, however, is a far more serious failure: discharging an infectious, symptomatic patient back into the general population fuels community spread and delays critical, life-saving interventions such as supplemental oxygen or antivirals.

The primary objective of any medical screening AI is to maximize sensitivity (recall). The model achieved a false-negative count of only 3 out of 200 positive cases, a false-negative rate (FNR) of 1.5%, indicating strong reliability for a clinical safety-critical screening role.

V. DISCUSSION, ETHICAL IMPLICATIONS, AND DEPLOYMENT

This project moves beyond a theoretical, textbook re-implementation of a neural network and directly addresses the logistical challenges of applying AI in modern healthcare settings.

A. Architectural and Computational Innovation

The primary innovation of this work lies in its structural feature-fusion mechanism. Standard CNNs act as black boxes that memorize the easiest path to minimize loss, which can involve memorizing a hospital’s scanner-specific watermark or border crop rather than the underlying pathology. By explicitly forcing the network to simultaneously solve an autoencoder reconstruction task, it is structurally discouraged from ignoring the true anatomical structure of the lungs — the reconstruction loss $\mathcal{L}_{recon}$ acts as a continuous regularizing signal throughout training, not merely a pretraining step, since both branches are optimized jointly at every gradient update.

Scalability was also a primary engineering constraint. Table III contrasts the parameter count and approximate inference cost of the chosen ResNet18 backbone against deeper alternatives. Using ResNet18 (approximately 11 million parameters) rather than a bulkier ResNet152 (approximately 60 million parameters) reduces inference time to a few milliseconds per scan, allowing the model to run efficiently on standard, low-end hospital hardware without requiring expensive GPU clusters — an important consideration for deployment in resource-constrained clinical settings, including secondary and rural care centers.

TABLE III
BACKBONE PARAMETER COUNT AND RELATIVE INFERENCE COST

Backbone	Params	Rel. Cost	GPU Needed
ResNet18 (used)	$\approx 11\text{M}$	$1\times$	Not required
ResNet50	$\approx 25\text{M}$	$\approx 2.3\times$	Recommended
ResNet152	$\approx 60\text{M}$	$\approx 5.5\times$	Required

B. Ethical Considerations and Dataset Bias

A persistent issue in medical AI is dataset bias: models trained exclusively on data from one demographic or geographic region often fail when deployed globally. While the Kaggle dataset used in this study is reasonably robust and class-balanced, true clinical deployment requires rigorous multi-center validation across scanner manufacturers, acquisition protocols, and patient populations before regulatory approval could be considered. Handling of medical imagery also requires strict adherence to privacy regulations (e.g., HIPAA in the United States, or equivalent regional frameworks); all DICOM files must be fully anonymized, stripping patient identifiers and embedded metadata, before being processed by the AI pipeline. The structural regularization provided by the autoencoder specifically helps mitigate geographic and scanner-level bias by encouraging the network to rely on universal human anatomy rather than hospital-specific digital signatures, but it does not eliminate the need for prospective clinical validation, human-in-the-loop review, or a documented model card disclosing training data provenance and known limitations.

C. Deployment Readiness

A highly accurate model has little clinical value if confined to a script that medical professionals cannot operate. The final trained weights were wrapped and deployed into a Streamlit web application (`app.py`) with a clean, intuitive graphical interface: a clinician can drag and drop a raw CT scan into the browser, and the backend immediately applies the necessary normalizations and tensor transformations, passes the image through the inference graph, and returns an instantaneous diagnostic prediction along with a mathematical confidence percentage. This closes the loop between model training and bedside usability and demonstrates that the system

is deployment-ready for pilot clinical use, subject to the validation caveats noted above.

VI. CONCLUSION

A. Concluding Remarks

The proposed hybrid pipeline, fusing a convolutional autoencoder with a transfer-learned ResNet18 architecture, is a highly effective, automated solution for detecting COVID-19 from thoracic CT scans. By marrying unsupervised structural regularization with rapid, supervised textural classification, the model achieves state-of-the-art accuracy (99.0%) while mitigating the domain-shift overfitting that pervades standard deep learning applications, resulting in an exceptionally low false-negative rate of 1.5%. The system is computationally efficient, mindful of dataset bias, fully deployable via a web interface, and shows strong potential as a life-saving clinical assistive tool in radiological screening.

B. Future Work

Three directions are planned for immediate expansion:

- 1) **Explainable AI (XAI):** integrating Grad-CAM (Gradient-weighted Class Activation Mapping), which traces classification-decision gradients back to the final convolutional layer to generate a heatmap highlighting the exact ground-glass opacities driving each diagnosis, increasing clinician trust.
- 2) **Multi-class classification:** extending the current binary formulation (COVID vs. normal) to distinguish COVID-19, viral pneumonia, bacterial pneumonia, and healthy tissue.
- 3) **Vision Transformers (ViTs):** evaluating self-attention-based architectures that view the entire image globally rather than through local convolutional windows, potentially yielding greater robustness to spatial translation and scanner noise.

CODE AVAILABILITY

The complete source code and deployment application are publicly available at https://github.com/a-abuzar/DL_CT-Based-COVID-19-Detection-via-Autoencoders-and-CNNs_727824tuam004.

REFERENCES

- [1] X. Ouyang *et al.*, “Dual-Sampling Attention Network for Diagnosis of COVID-19 from Community Acquired Pneumonia,” *IEEE Trans. Med. Imag.*, vol. 39, no. 8, pp. 2595–2605, 2020.
- [2] H. Gunraj, L. Wang, and A. Wong, “COVID-Net CT-2: A Tailored Deep Convolutional Neural Network Design for Detection of COVID-19 Cases from Chest CT Images,” *Frontiers in Medicine*, vol. 7, 2020.
- [3] T. Ozturk *et al.*, “Automated Detection of COVID-19 Cases Using Deep Neural Networks with X-ray Images,” *Comput. Biol. Med.*, vol. 121, 2020.
- [4] A. M. Hasan *et al.*, “Anomaly Detection-Based Deep Learning Approaches for COVID-19,” *Informatics in Medicine Unlocked*, vol. 24, 2021.
- [5] P. Silva *et al.*, “Auto-Encoder-Based Feature Extraction for COVID-19 CT Scan Classification,” *Pattern Recognition Letters*, 2022.